Ericsson PE Node WFD-PE-02 QoS Congestion Event Resolution SOP

Node: WFD-PE-02 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_5395

Description

This Standard Operating Procedure (SOP) is designed to resolve Critical severity QoS congestion events (Microbursts, interface tail drops, QoS queue starvation) on Ericsson vendor equipment, specifically on the Whitefield POP site's Node ID WFD-PE-02, which operates as a Provider Edge (PE) node. The potential failure domain of QoS congestion events on this specific role and vendor equipment is largely attributed to the intricacies of the Label Switched Path (LSP) and its associated Label Edge Routers (LERs). The LSP is a critical component in the setup of MPLS networks, as it defines the path a data packet will take through the network. Misconfigurations or anomalies in the LSP can lead to congestion events, such as microbursts, which are characterized by brief periods of high packet loss and high network utilization. Other potential causes include interface tail drops, which occur when the network interface is overwhelmed with incoming packets and cannot process them in a timely manner, resulting in packet loss. Additionally, QoS queue starvation can occur when a network device's QoS scheduler is unable to allocate sufficient bandwidth to a particular flow, leading to packet loss and congestion. To resolve these issues, it is essential to carefully troubleshoot the network configuration, verify the LSP path, and analyze the QoS queue configuration to identify and address any misconfigurations or anomalies.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Whitefield POP site to provide physical access to the equipment, if necessary.

- Verify the status of the redundant links to ensure they are functioning correctly and can handle the increased traffic load.

- Review the network configuration to identify any potential misconfigurations or anomalies that may be contributing to the QoS congestion event.

Potential Root Causes

Cause	Probability
Misconfigured LSP or its associated LERs	High
Interface Overload or misconfigured interface settings	Medium
QoS queue starvation or misconfigured QoS scheduler	Low

Diagnostic Commands & Expected Outputs

```
Command: show mpls ldp database, Expected Output: Displays the LDP database, including the LSPs and their associated LERs., Description:
```

```
This command is used to verify the LDP database and identify any
potential misconfigurations or anomalies in the LSP setup.

Command: show interface statistics, Expected Output: Displays the
interface statistics, including the number of packets sent and
received, as well as any errors or drops., Description: This command
is used to analyze the interface statistics and identify any potential
issues, such as interface overload or misconfigured interface
settings.

Command: show qos queue statistics, Expected Output: Displays the QoS
queue statistics, including the number of packets sent and received,
as well as any errors or drops., Description: This command is used to
analyze the QoS queue statistics and identify any potential issues,
such as QoS queue starvation or misconfigured QoS scheduler.
```

Resolution Steps

Step 1: Step: Verify and correct the LSP or its associated LERs, Command: configure replace mpls ldp database, Description: This step involves verifying the LDP database and correcting any potential misconfigurations or anomalies in the LSP setup. The command configure replace mpls ldp database is used to replace the LDP database with a corrected version.

Step 2: Step: Verify and correct the interface settings, Command: configure replace interface, Description: This step involves verifying the interface settings and correcting any potential issues, such as interface overload or misconfigured interface settings. The command configure replace interface is used to replace the interface settings with a corrected version.

Step 3: Step: Verify and correct the QoS queue configuration, Command: configure replace qos queue, Description: This step involves verifying the QoS queue configuration and correcting any potential issues, such as QoS queue starvation or misconfigured QoS scheduler. The command configure replace qos queue is used to replace the QoS queue configuration with a corrected version.

Post-Check Verification

```
Command: show mpls ldp database, Expected Output: Displays the LDP
database, including the LSPs and their associated LERs., Description:
This command is used to verify the LDP database and ensure that the
LSP setup is correct.

Command: show interface statistics, Expected Output: Displays the
interface statistics, including the number of packets sent and
received, as well as any errors or drops., Description: This command
is used to analyze the interface statistics and ensure that the
interface settings are correct.

Command: show qos queue statistics, Expected Output: Displays the QoS
queue statistics, including the number of packets sent and received,
as well as any errors or drops., Description: This command is used to
analyze the QoS queue statistics and ensure that the QoS queue
configuration is correct.
```

Roll-back Procedures

- Step: Restore the original LSP or its associated LERs, Command: configure rollback mpls ldp database, Description: This step involves restoring the original LDP database to prevent any potential disruptions to the network.

- Step: Restore the original interface settings, Command: configure rollback interface, Description: This step involves restoring the original interface settings to prevent any potential disruptions to the network.

- Step: Restore the original QoS queue configuration, Command: configure rollback qos queue, Description: This step involves restoring the original QoS queue configuration to prevent any potential

disruptions to the network.